

TestYourself AI — Product Description

TestYourself AI is an advanced, high-performance Telegram companion bot designed for students, educators, and professionals to streamline study workflows. By leveraging modern Retrieval-Augmented Generation (RAG) and the Gemini Pro API, users can transform any raw academic or technical PDF document into an interactive, structured learning sandbox.

Core Features

1. Seamless Document Indexing

- **Robust PDF Parsing:** Automatically reads and indexes uploaded PDF documents of any size.
- **Smart Partitioning:** Automatically handles documents yielding 100+ text chunks by partitioning them into batch segments of 100 before fetching Gemini embeddings, successfully bypassing API batch limits.
- **Deterministic Vectors:** Maps document content using `text-embedding-004` and stores them in high-speed, local FAISS vector databases.

2. Context-Aware RAG Dialogue

- **Interactive Q&A:** Chat naturally with your document to clarify concepts, locate information, or summarize sections.
- **Exact Page Citations:** Every RAG reply includes exact references (e.g., `[References: Page(s) 12, 15]`) so you can verify facts in the source document.

3. Dynamic Study Material Generation

Generate highly accurate study material on-demand. All generations are styled with **Zero-Temperature Determinism (0.0)** to enforce absolute factual accuracy and eliminate duplicate questions:

- **MCQ Quizzes:** Generates dynamic multiple-choice quizzes with a promptable question limit (from 5 up to 50 questions).
- **Interactive Flashcards:** Curates active-recall decks displaying front/back study prompts.
- **Conceptual Q&A Sheets:** Summarizes the document's core pillars into plain-text question-and-answer pairs.
- **Interview Training Guides:** Formulates realistic interview questions paired with ideal target responses.

4. Styled PDF Document Exporters

Export any generated study sheet into a clean, beautifully formatted PDF document instantly:

- **"pdf" Chat Trigger:** Type `"pdf"` in the chat at any time to pull up a custom download menu listing all compiled sheets.
- **Inline Shortcut Buttons:** Direct  `Download PDF` buttons are embedded directly onto the quiz results slide and study sheet generation slides.
- **Unicode Cleaning:** Automatically sanitizes emojis and unsupported Unicode formatting to prevent PDF compilation crashes under ReportLab.

5. High-Concurrency Architecture

Optimized to run smoothly with **20+ concurrent active users**:

- **Concurrent Update Handlers:** Leverages concurrent asyncio task workers to process commands and callbacks simultaneously.
- **CPU & I/O Thread Offloading:** CPU-bound operations (PDF saving, text parsing, FAISS file reads, and JSON serialization) are executed inside background threads via `asyncio.to_thread`, keeping the event loop fully responsive.
- **Index Cache:** Caches active vector databases in RAM to eliminate disk I/O bottlenecks.

🔧 How to Launch

1. **Environment Setup:** Ensure your `.env` contains:

```
env
TELEGRAM_BOT_TOKEN=your_telegram_bot_token
GEMINI_API_KEY=your_gemini_api_key
DATABASE_URL=sqlite+aiosqlite:///./stark.db
```

2. **Start the Backend Server:**

```
bash
python -m src.backend.main
```

3. **Start the Telegram Bot:**

```
bash
python -m src.bot.main
```